

AR 525: Reinforcement Learning for Robotics

Course Code: AR 525

Course Name: Reinforcement Learning for Robotics

L-T-P-C: 3-0-0-3

Prerequisites: IC 272: Machine Learning / CS 305 or CS 362: Artificial Intelligence / Equivalent courses and Programming Knowledge (Python)

Intended for: B.Tech, M.Tech, M.Tech. (R), PhD

Mutual Exclusion: None

Approval: 48th Senate (listed in the B.Tech General Engineering curriculum); syllabus approved in 64th BoA

Preamble

Reinforcement Learning (RL) is a paradigm that models trial-and-error decision making, enabling physical robotic systems to learn optimal behaviors through interaction with uncertain, dynamic environments where explicit supervision is unavailable. This course introduces the mathematical foundations of RL while emphasizing robotics-specific challenges, including continuous state-action spaces, partial observability, contact-rich dynamics, safety constraints, sim-to-real transfer, and data efficiency. Core algorithms will be discussed alongside robot-centric case studies such as manipulation, locomotion, mobile navigation, and human-robot interaction. Applications will be drawn primarily from robotics, with additional examples from healthcare, autonomous systems, and human-AI collaboration.

Course Modules

- **Introduction to Reinforcement Learning:** RL problem formulation with robotic agents, Basic Definitions, Examples from robotic navigation, manipulation, and control, Strengths and limitations of RL for real-world robots (sample efficiency, safety, interpretability) (2 hours)
- **Finite Markov Decision Process (MDP):** Markov processes and MDPs with robot state representations, Continuous vs. discrete state/action spaces in robotics, Rewards for robotic tasks (sparse vs. shaped rewards), Partially Observable MDPs (POMDPs) for sensing-limited robots, Value functions and optimality in motion planning and control, Approximation issues arising from high-dimensional robot states (6 hours)
- **Dynamic Programming for Robotics (Planning by DP):** Policy and value iteration in robot motion planning, Relationship to shortest-path and optimal control problems. Generalized Policy Iteration for robotic decision-making, Limitations of DP for high-dimensional robotic systems, Convergence and contraction mappings with robotics examples (6 hours)
- **Model-Free Prediction in Robotic Systems:** Monte Carlo and TD learning for robotic task evaluation, Eligibility traces for delayed rewards in manipulation tasks, Temporal credit assignment in sequential robot actions, On-policy vs. off-policy prediction in physical systems (4 hours)
- **Model-free Control for Robotics:** On-policy and off-policy control for robots, Exploration-exploitation trade-offs in real robots, Safety-aware exploration strategies (4 hours)
- **Value Function Approximation for High-Dimensional Robots:** Function approximation for continuous robotic states, Deep Q-Learning for robotic decision-making, Experience replay and sample reuse in robotics, Challenges of instability and overestimation in control tasks (4 hours)

- **Policy Gradient Methods for Robot Control:** Stochastic policies for continuous control, Actor–critic methods for robotic manipulation and locomotion, Comparison with classical optimal control methods, Variance reduction techniques in robot learning (4 hours)
- **Integrating Planning and Learning in Robotics:** Model-based RL for robotic dynamics, Learning residual models over analytical physics, Simulation-based planning and rollouts, Combining RL with trajectory optimization and MPC (4 hours)
- **Deep Reinforcement Learning for Robotics:** Continuous-control algorithms: PPO, DDPG, TD3, SAC, TRPO, GRPO, Stability and robustness issues in physical robots, Sim-to-real transfer, domain randomization (4 hours)
- **Reinforcement Learning with Human Feedback in Robotics (Coaching):** Imitation learning from human demonstrations, Inverse RL for learning reward functions in robotics, Human preference learning for shared autonomy, Safety, alignment, and trust in human–robot interaction (4 hours)

Text Books

1. Richard S. Sutton and Andrew G. Barto; *Reinforcement Learning: An Introduction*; 2nd Edition, MIT Press, 2020.
2. Bertsekas, D. (2019). *Reinforcement learning and optimal control* (Vol. 1). Athena Scientific.

References

1. Ghasemi, M., Moosavi, A. H., & Ebrahimi, D. (2024). *A comprehensive survey of reinforcement learning: From algorithms to practical challenges*. arXiv preprint arXiv:2411.18892.
2. Hausknecht, M., Stone, P., & Mc, O. P. (2016, July). *On-policy vs. off-policy updates for deep reinforcement learning*. In Deep reinforcement learning: frontiers and challenges, IJCAI 2016 Workshop. AAAI Press New York, NY, USA.
3. Grondman, I., Busoniu, L., Lopes, G. A., & Babuska, R. (2012). *A survey of actor-critic reinforcement learning: Standard and natural policy gradients*. IEEE Transactions on Systems, Man, and Cybernetics, part C (applications and reviews), 42(6), 1291-1307.
4. Moerland, T. M., Broekens, J., Plaat, A., & Jonker, C. M. (2023). *Model-based reinforcement learning: A survey*. Foundations and Trends® in Machine Learning, 16(1), 1-118.
5. Lillicrap, T. P., Hunt, J. J., Pritzel, A., Heess, N., Erez, T., Tassa, Y., ... & Wierstra, D. (2015). *Continuous control with deep reinforcement learning*. arXiv preprint arXiv:1509.02971.
6. Raina, D., Balakuntala, M. V., Kim, B. W., Wachs, J., & Voyles, R. (2024). *Coaching a robotic sonographer: learning robotic ultrasound with sparse expert's feedback*. IEEE Transactions on Medical Robotics and Bionics.